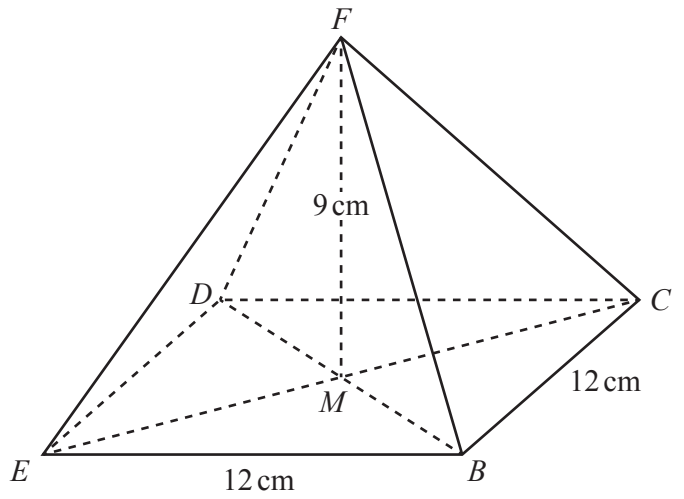
0580_m24_qp_42_Q4

4 (a)



NOT TO SCALE

The diagram shows a pyramid with a square base *BCDE*.
The diagonals *CE* and *BD* intersect at *M*, and the vertex *F* is directly above *M*.
BE = 12 cm and *FM* = 9 cm.

(i) Calculate the volume of the pyramid.

[The volume, *V*, of a pyramid with base area *A* and height *h* is $V = \frac{1}{3}Ah$.]

..... cm^3 [2]

(ii) Calculate the total surface area of the pyramid.

..... cm^2 [5]

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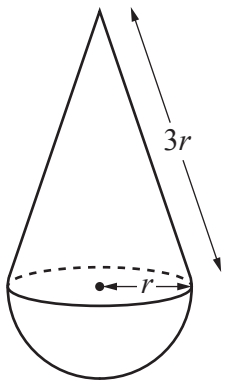
(b)



NOT TO SCALE

The diagram shows a toy made from a cone and a hemisphere.
The base radius of the cone and the radius of the hemisphere are both r cm.
The slant height of the cone is $3r$ cm.

The total surface area of the toy is $304\,\text{cm}^2$.

Calculate the value of r.

[The curved surface area, A, of a cone with radius r and slant height l is $A = \pi r l$.]
[The curved surface area, A, of a sphere with radius r is $A = 4\pi r^2$.]

$r =$ ..... [4]